

REFERRAL — RAVENSGATE, 19 NOVEMBER

SITE	RAVENSGATE DAM	OPERATOR	CAIRNHOLD POWER	DEATHS	2
POLICE FINDING	MURDER-SUICIDE	RDSI POSITION	DISSENT	AUTHOR	CHIEF INSPECTOR R. HALSEY

The Inspectorate does not investigate homicide and does not presume to. We investigate why plant kills people. On this occasion the two enquiries have produced findings that cannot both be correct, and the disagreement is not a matter of judgement. It is a matter of water.

The police investigation concluded that Ines Ballard killed Aurelio Sarn at 06:03 and then entered the forebay and drowned herself. Her body was recovered at 07:25 from **trash rack panel 4, inside the intake gallery**, at elevation 218.4 m.

THE POINT AT ISSUE

The intake gallery is not open water. It is a sealed concrete structure downstream of intake gate 2. It is dry above the sill whenever that gate is shut, and intake gate 2 was shut from 03:12:44 that morning until 08:40:19.

The police report does not address how the deceased came to be inside it. It records where she was found and does not ask how she got there. **We are unable to identify any route by which a body could enter the intake gallery after 03:12.**

We make no allegation and offer no alternative account. We say only that a finding which places a living woman at a switchgear cabinet at 06:02 must also explain how her body reached a location that was, by then, unreachable.

The Inspectorate's technical file is provided in full. Our own conduct in respect of the 2019 instrumentation renewal at this site is also within the scope of this review, and we have not redacted it.

SIGNED

R. HALSEY, CHIEF INSPECTOR

DATE

14 JAN

DEATHS OF A. SARN AND I. BALLARD

OPERATION	OP 24-11-2233	OPENED	19 NOV	CLOSED	08 DEC (19 DAYS)
SIO	DCI G. NAUNTON	FINDING	MURDER-SUICIDE	PERSONS ON SITE	6

SUMMARY OF FINDING

That Ines Ballard, Control Engineer, caused the death of Aurelio Sarn, Station Manager, by deliberately energising switchgear on which he was working, and thereafter took her own life by entering the forebay.

THE THREE LEGS

#	BASIS	SOURCE
1	Her credential. Cabinet C-14 was opened at 05:47:31 on credential I.BALLARD. Feeder tie F-3 was closed at 06:02:14 on credential I.BALLARD. Mr Sarn died at 06:03.	D-1
2	The argument. Ms Ballard and Mr Sarn were overheard at 02:25 in a serious disagreement in or adjacent to the control room. The subject was not established.	witness
3	No third party. The site was sealed by landslip from 19:40. Of the four other persons present, each was engaged in documented work at the material time, and none had reason to harm either deceased.	rosters, logs

MATTERS NOT RESOLVED

Recorded here for completeness at the direction of the SIO.

- No note or communication of intent was recovered from Ms Ballard.
- Ms Ballard's credential card was not on her person when recovered. Its location is recorded in the property schedule and was not thought material.
- The Office of the Chief Medical Examiner was unable to estimate a time of death for Ms Ballard. The investigation proceeded on the basis that she died after Mr Sarn, that being the sequence the credential evidence implies.
- No motive was established. Professional friction between the deceased was reported by two witnesses and denied by two others.

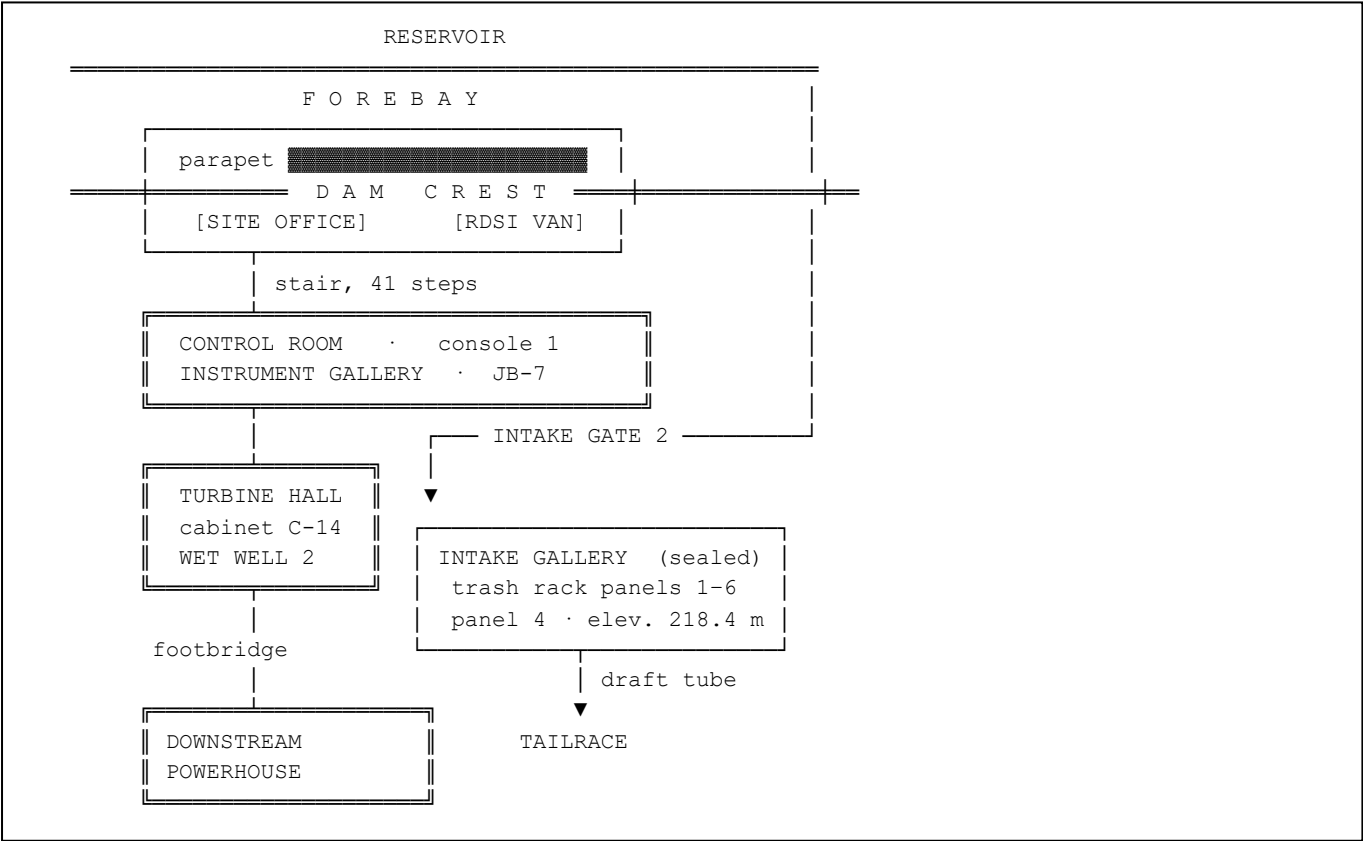
Note by the Case Review Unit. The third bullet above is the whole of the reason this file is in front of you. The sequence of the two deaths was not established by evidence. It was inferred from the finding, and the finding was then built on the sequence.

RAVENSGATE — ORIENTATION

PERSONS ON SITE, 18-19 NOVEMBER

NAME	POST	WHERE THEY WERE WORKING
SARN, Aurelio (58)	Station Manager	Came in for the storm. Mobile across the site. DECEASED 06:03
BALLARD, Ines (41)	Control Engineer	Control room and instrument gallery. DECEASED — time unknown
VEY, Halloran (47)	Shift Control Engineer	Console 1, control room
OKONKWO, Dessa (34)	Mechanical Maintenance Technician	Wet well 2, pump seal repair
WRENN, Piotr (61)	Dam Safety & Instrumentation Officer, RDSI	Instrument van, dam crest. On site for a scheduled inspection; stranded by the landslip.
MBEKI, Cato (26)	Assistant Operator	Downstream powerhouse

SITE PLAN — SCHEMATIC



WALKING TIMES — MEASURED BY THE REVIEW UNIT

FROM	TO	MINUTES
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Control room	Forebay parapet	4
Control room	Site office (across the crest)	5
Control room	Instrument gallery (41 steps)	3
Control room	Turbine hall, cabinet C-14	6
Downstream powerhouse	Anywhere on the main site	5 – footbridge only; service tunnel gated and padlocked for the storm

STORM CHRONOLOGY — UNCONTESTED

18 NOV 17:00 Storm Bettany. Night complement on site.

18 NOV 19:40 **Landslip severs the access road.** No person entered or left the site until 14:20 the following afternoon.

19 NOV Peak inflow to the reservoir.
04:00–05:00

19 NOV 06:10 Mr Sarn found in the turbine hall.

19 NOV 07:25 Ms Ballard found in the intake gallery.

POST-MORTEM — SARN, AURELIO

CASE	OCME 24-2871	AGE / SEX	58 / M	EXAMINER	DR H. ADEYEMI
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EXTERNAL EXAMINATION

Full-thickness contact burn to the right palm, 34 mm, with central charring and a raised margin — a classic electrical entry wound. Corresponding exit wound to the left forearm, 21 mm, at the point of contact with an earthed steel cabinet frame. Singeing of the forearm hair. No other injury of any kind. No defensive wounds. Nothing to suggest a struggle, and nothing to suggest restraint.

INTERNAL EXAMINATION

Findings consistent with a **hand-to-hand current path across the thorax** — the configuration that reliably arrests the heart. Contraction-band necrosis of the myocardium. No pre-existing cardiac disease of significance. No other contributory pathology.

TOXICOLOGY

Negative throughout. No alcohol, no drugs of abuse, no prescribed medication above therapeutic range, no carbon monoxide.

CAUSE OF DEATH

Ia. ELECTROCUTION — consistent with contact with an 11 kV conductor.

MANNER: UNLAWFUL KILLING. Death was effectively instantaneous.

TIME OF DEATH

06:03, 19 November. This is not a pathological estimate and does not need to be. The earth-fault alarm cascade at the station is timestamped 06:03, the deceased was found at 06:10, and the body was warm and unfixed on examination at the scene. **The time of this death is not in dispute and never has been.**

POST-MORTEM — BALLARD, INES

CASE	OCME 24-2872	AGE / SEX	41 / F	EXAMINER	DR H. ADEYEMI
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EXTERNAL EXAMINATION

Post-mortem abrasion and contusion to the anterior trunk, shoulders and knees, distributed in the pattern produced by prolonged contact with a coarse steel grating under flow. No injury shows a vital reaction — **all of it occurred after death.**

No defensive injuries. No grip marks to the arms. No ligature. No injury to the back of the head or neck. Fingernails intact and clean.

INTERNAL EXAMINATION

Frothy oedema fluid in the airway. Over-inflated lungs with rib impressions. Water in the stomach and duodenum. **Diatom recovery from bone marrow matches the forebay profile** and not the tailrace. She was alive when she entered the water and she entered it upstream.

CAUSE OF DEATH

Ia. DROWNING — freshwater, cold.

MANNER: OPEN. The pathology cannot distinguish between entering the water voluntarily, falling, and being put in. There is nothing on this body that decides it.

TIME OF DEATH — STATEMENT OF LIMITATION**READ THIS BEFORE RELYING ON ANY ESTIMATE**

The decedent was immersed in moving water at **3–5 °C** for an unknown but substantial period. This invalidates, individually and together:

Algor mortis	Body temperature equilibrated with the water long before recovery. The differential carries no information.
Rigor mortis	Cold profoundly retards onset and the body was additionally subject to continuous mechanical loading against the rack.
Livor mortis	Position was not fixed and flow was continuous. No usable pattern.
Vitreous potassium	Unreliable below 10 °C ambient. This Office does not report it in immersion cases.

THIS OFFICE CANNOT ESTIMATE A TIME OF DEATH FOR THIS DECEDENT AND DECLINES TO ATTEMPT ONE. Any sequence of the two deaths must be established from evidence other than pathology. It cannot be established from this report, and this report should not be cited as if it had been.

SCADA CORE EXTRACT

Master station log, 18 NOV 17:00 – 19 NOV 09:00. Time source: GPS-disciplined, synchronised throughout. Extract certified complete for the systems listed. Other subsystems (console session, alarm acknowledgement, permit and access) are held separately.

1 · SWITCHING & ISOLATION — TURBINE HALL

TIMESTAMP	EVENT	CREDENTIAL	INITIATED AT
19 NOV 05:47:31	CABINET C-14 — DOOR OPENED	I.BALLARD	LOCAL (cabinet)
19 NOV 05:52:06	FEEDER F-5 — ISOLATOR OPENED, LOTO PADLOCK APPLIED	A.SARN	LOCAL (switchroom)
19 NOV 05:53:44	FEEDER F-5 — PROVING TEST, DEAD CONFIRMED	A.SARN	LOCAL
19 NOV 06:02:14	FEEDER F-3 — TIE CLOSED	I.BALLARD	CONTROL ROOM, CONSOLE 1
19 NOV 06:03:02	EARTH FAULT — C-14 · CASCADE, 11 ALARMS	—	—
19 NOV 06:03:19	ALL 11 ALARMS ACKNOWLEDGED	H.VEY	CONTROL ROOM, CONSOLE 1

SYSTEMS NOTE SUPPLIED BY CAIRNHOLD POWER

Mr Sarn isolated feeder F-5, applied his personal lock, and proved the circuit dead. He did this correctly and the log records it. He was killed by feeder F-3.

2 · GATE OPERATIONS

TIMESTAMP	GATE	EVENT	INITIATED AT
19 NOV 02:31:08	INTAKE GATE 2	OPEN	CONTROL ROOM, CONSOLE 1
19 NOV 03:12:44	INTAKE GATE 2	CLOSED	CONTROL ROOM, CONSOLE 1
19 NOV 08:40:19	INTAKE GATE 2	OPEN (recovery operation)	CONTROL ROOM

Intake gates 1 and 3: not operated at any point during the storm.

Spillway gates: not operated. Overflow was passive throughout.

3 · CREDENTIAL EVENTS — ALL SYSTEMS, 19 NOV 00:00–07:00

TIMESTAMP	CREDENTIAL	EVENT
01:50:12	P.WRENN	Control room – entry
01:58:40	P.WRENN	Control room – exit
02:12:05	D.OKONKWO	Confined space permit 1144 – entry, wet well 2
03:40:22	D.OKONKWO	Confined space permit 1144 – exit
05:35:41	D.OKONKWO	Confined space permit 1145 – entry, wet well 2
05:44:03	H.VEY	Control room – exit
05:46:12	H.VEY	Control room – entry
05:47:31	I.BALLARD	Cabinet C-14 – door opened
06:02:14	I.BALLARD	Feeder F-3 tie – closed, at console 1
06:05:51	C.MBEKI	Footbridge – crossing to main site
06:25:30	D.OKONKWO	Confined space permit 1145 – exit
No credential event of any kind attributed to I.BALLARD between 19 NOV 01:12 and 05:47:31.		

4 · RESERVOIR LEVEL & INFLOW

TIME	LEVEL M AOD	INFLOW M ³ /S	NOTE
19 NOV 00:00	241.10	88	Rising
19 NOV 02:00	241.44	119	Rising
19 NOV 02:31	241.52	126	Gate 2 opened
19 NOV 03:12	241.49	131	Gate 2 closed
19 NOV 04:12	241.71	184	Peak inflow
19 NOV 05:00	241.83	160	Falling
19 NOV 07:00	241.90	102	Falling

5 · SYSTEMS HELD SEPARATELY

Not included in this extract. Each is available to the review on request.

SYSTEM	HOLDER
Console 1 — alarm acknowledgement log	Cairnhold operations
Trash rack differential-pressure trace	Cairnhold civil
Property and credential audit (scene)	County Constabulary
As-built schedule amendment history	Cairnhold engineering records
Contract RG-0419 file, and payments thereunder	Cairnhold procurement